

15 Jan

Linearization

$$\begin{aligned} \dot{x} &= f(x) && \text{time-invariant} \\ \bar{x} &: \text{a solution (feasible trajectory)} \\ \downarrow & \\ x(t) &= \bar{x}(t) + \tilde{x}(t) && \left| \begin{array}{l} \tilde{x}(t) = x(t) - \bar{x}(t) \\ \text{perturbation around } \bar{x} \end{array} \right. \\ \uparrow & \\ & \text{given trajectory} \\ & \text{e.g. an equilibrium} \\ \text{derivative} & \left(\begin{array}{l} \dot{x}(t) = \dot{\bar{x}}(t) + \dot{\tilde{x}}(t) \Rightarrow \dot{\tilde{x}}(t) = \dot{x}(t) - \dot{\bar{x}}(t) \end{array} \right. \end{aligned}$$

Because $\bar{x}(t)$ is a solution

$$\Rightarrow \dot{\bar{x}}(t) = f(\bar{x}(t))$$

$$\dot{\tilde{x}}(t) = f(x) - f(\bar{x})$$

$$\boxed{\dot{\tilde{x}}(t) = f(\bar{x} + \tilde{x}) - f(\bar{x})} \quad \text{dynamics of the perturbed variable}$$

Recall f is time invariant

• $\bar{x}$ is an eq.

$$\dot{\tilde{x}}(t) = f(\bar{x} + \tilde{x}) - \cancel{f(\bar{x})}^0$$

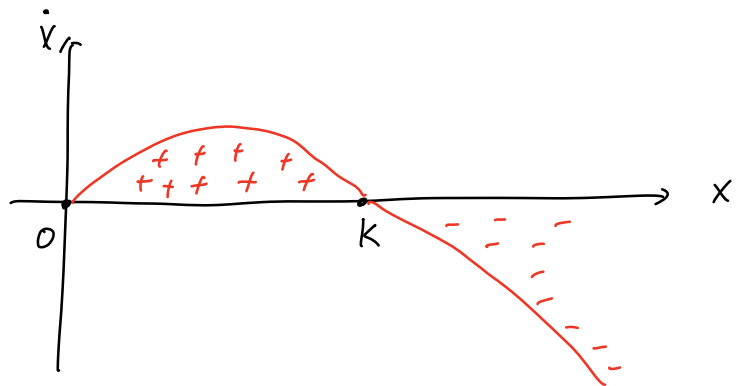
time-invariant

• $\bar{x}$ is not an eq.

$$\dot{\tilde{x}} = f(\bar{x}(t) + \tilde{x}) - f(\bar{x}(t))$$

time-varying

• No Approximation



$$\dot{\tilde{x}} = f(\bar{x} + \tilde{x}) - f(\bar{x})$$

For sufficiently small $\tilde{x}$

$$f(\bar{x} + \tilde{x}) - f(\bar{x}) \approx \left. \frac{\partial f}{\partial x} \right|_{x=\bar{x}} \cdot \tilde{x} \quad \text{Taylor Approximation}$$

$\Downarrow$

Approximate dynamics for the perturbed variable:

$$\dot{\tilde{x}} = \left(\left. \frac{\partial f}{\partial x} \right|_{x=\bar{x}} \right) \tilde{x}$$

A

$$\dot{\tilde{x}} = A \tilde{x}$$

$$\Rightarrow \boxed{\tilde{x}(t) = e^{At} \tilde{x}(0)} \quad \text{General solution}$$

$$e^{At} = I + At + \frac{A^2 t^2}{2!} + \frac{A^3 t^3}{3!} + \dots$$

Example: $\dot{x} = \underbrace{\alpha \left(1 - \frac{x}{k}\right)}_{f(x)} x$

$$\begin{aligned} \frac{\partial f}{\partial x} &= \frac{\partial}{\partial x} \left[\alpha \left(1 - \frac{x}{k}\right) x \right] \\ &= \alpha \left(1 - \frac{2x}{k}\right) \end{aligned}$$

• $\bar{x} = 0$

$$\left. \frac{\partial f}{\partial x} \right|_{x=\bar{x}=0} = \alpha$$

$$\tilde{\dot{x}} = \alpha \tilde{x}$$

$$\tilde{x}(t) = \tilde{x}(0) e^{\alpha t}$$

$$x = \tilde{x} + \bar{x} = \tilde{x}$$

$$\rightarrow x(t) = x(0) e^{\alpha t}$$

• $\bar{x} = k$

$$\left. \frac{\partial f}{\partial x} \right|_{x=\bar{x}=k} = -\alpha$$

$$\tilde{\dot{x}} = -\alpha \tilde{x}$$

$$\tilde{x}(t) = e^{-\alpha \tilde{x}(t)}$$

$$x = \tilde{x} + \bar{x} \rightarrow \tilde{x} = x - k$$

$$(x(t) - k) = (x(0) - k) e^{-\alpha t}$$

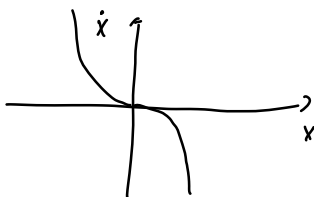
$$x(t) = k(1 - e^{-\alpha t}) + x(0) e^{-\alpha t}$$

$$= k + (x(0) - k) e^{-\alpha t}$$

• Small perturbations around $\bar{x} = 0$ are going to grow [$\bar{x} = 0$ is unstable]

• Small perturbation around $\bar{x} = k$ decay with time [$\bar{x} = k$ is locally asymptotically stable]

Ex. a) $\dot{x} = -x^3$

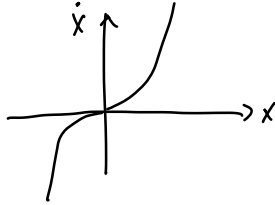


$$f(x) = -x^3$$

$$f(x) = 0 \Rightarrow x = 0$$

b) $\dot{x} = x^3$

$$f(x) = x^3$$



$$f(x)=0 \Rightarrow x=0$$

$$\bullet f = -x^3$$

$$\frac{\partial f}{\partial x} = -3x^2$$

$$\frac{\partial f}{\partial x} \Big|_{x=0} = 0$$

$$\bullet f = x^3$$

$$\frac{\partial f}{\partial x} = 3x^2$$

$$\frac{\partial f}{\partial x} \Big|_{x=0} = 0$$

$$\dot{\tilde{x}} = 0 \cdot \tilde{x} \quad (\text{both systems})$$

$$= 0$$

$$\therefore \tilde{x}(t) = \tilde{x}(0) \quad x = \tilde{x} + \overset{1}{x} 0$$

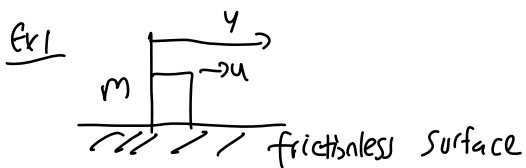
$$x(t) = x(0)$$

$$\dot{x}_1 = f_1(x) = f_1(x_1, x_2)$$

$$\dot{x}_2 = f_2(x) = f_2(x_1, x_2)$$

$$\frac{\partial f}{\partial x} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}$$

$$x(t) = e^{At} x(0)$$



$$m\ddot{y} = u$$

$$y = x_1$$

$$\dot{y} = \dot{x}_1 = x_2$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = \frac{u}{m}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$\begin{bmatrix} \dot{x}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}}_A \begin{bmatrix} x_1 \end{bmatrix} + \underbrace{\begin{bmatrix} y_m \end{bmatrix}}_B$$

$$\dot{X} = Ax + Bu$$

unforced system: $\dot{X} = Ax = f(X)$

$$= \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}_A X$$

$$X(t) = e^{At} X(0)$$

$$e^{At} = I + At + \frac{A^2 t^2}{2!} + \dots$$

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\Rightarrow x_2 = 0, x_1 \in \mathbb{R}$$

$$X=0 : \begin{array}{l|l} x_1=0 & x_1=1 \\ x_2=0 & x_2=0 \end{array} \quad \text{perturbation}$$

$$A^2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$e^{At} = I + At = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} t = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}$$

$$x_2(t) = x_2(0)$$

$$x_1(t) = x_1(0) + t x_2(0)$$

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\det(A - \lambda I) = \det \begin{pmatrix} -\lambda & 1 \\ 0 & -\lambda \end{pmatrix} = \lambda^2 = 0$$

$$\Rightarrow \lambda = 0$$

Ex2



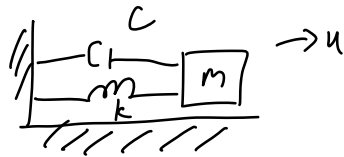
$$m\ddot{y} + ky = u$$

$$A = \begin{bmatrix} 0 & 1 \\ -k/m & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 1/m \end{bmatrix}$$

$$\det(A - \lambda I) = \det \begin{pmatrix} -\lambda & 1 \\ -k/m & -\lambda \end{pmatrix} = \lambda^2 + k/m = 0$$

$$\Rightarrow \lambda = \pm \sqrt{k/m} ;$$

Ex3



$$m\ddot{x}_2 + c\dot{x}_2 + kx_2 = u$$

$$y = k_1$$

$$x_2 = \dot{x}_1$$

$$\dot{x}_2 = 1/m (u - kx_1 - cx_2)$$

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -k/m & -c/m \end{bmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{bmatrix} 0 \\ 1/m \end{bmatrix}}_B u$$

$$\det(A - \lambda I) = \det \begin{pmatrix} -\lambda & 1 \\ k/m & -c/m - \lambda \end{pmatrix} = \lambda^2 + c/m \lambda + k/m = 0$$

$$\Rightarrow \lambda = \frac{-\frac{c}{m} \pm \sqrt{\frac{c^2}{m^2} - 4\frac{k}{m}}}{2}$$

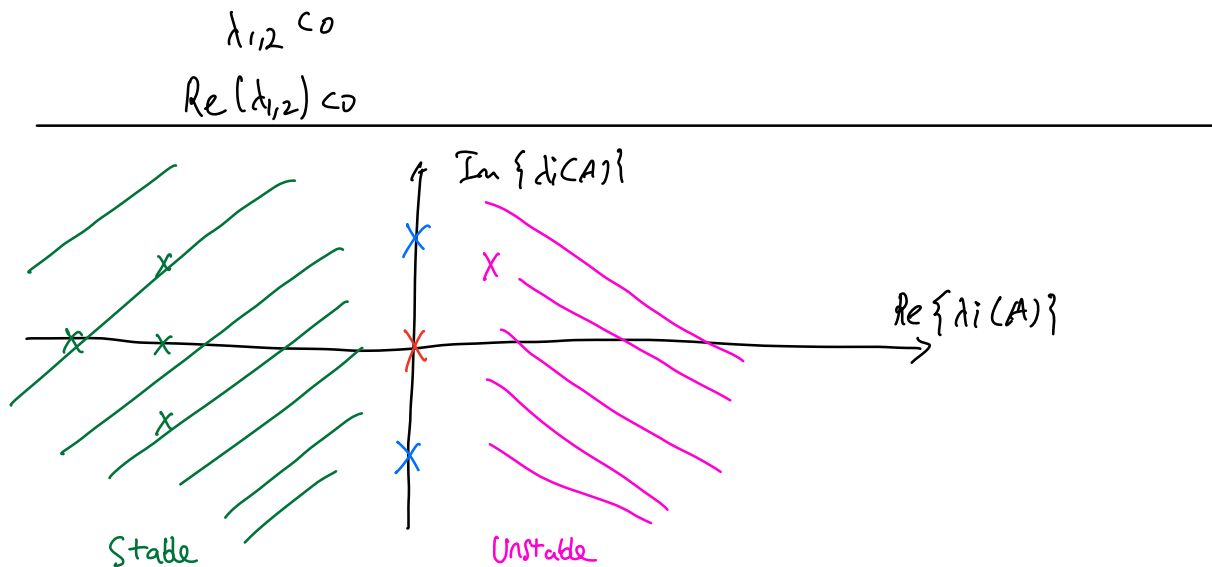
$$= \frac{-c \pm \sqrt{c^2 - 4km}}{2m}$$

$$\boxed{c > 2\sqrt{mk}}$$

$$\lambda^2 + \frac{c}{m}\lambda + \left(\frac{c}{2m}\right)^2 = \left(\frac{c}{2m}\right)^2 - \frac{k}{m}$$

$$\lambda + \frac{c}{2m} = \pm \sqrt{\frac{c^2}{4m^2} - \frac{k}{m}}$$

$$\lambda = \frac{-c \pm \sqrt{c^2 - 4km}}{2m}$$



→ All e-values strictly in the left half plane $\Rightarrow x=0$ is stable

("every trajectory converges to $x=0$ ")

At least one e-value in the right half plane $\Rightarrow x=0$ is not stable

("some trajectories grow unbounded")

2D Systems

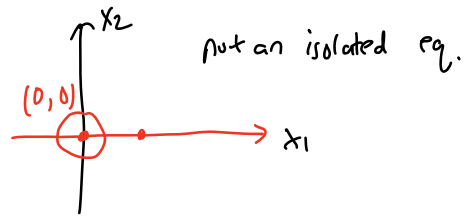
Characteristic polynomial: $\lambda^2 + a_1\lambda + a_0 = 0$

$a_1 > 0, a_0 > 0 \Rightarrow$ stability

Range of phenomena for $\dot{x} = f(x)$ (nonlinear)

- 1: Multiple "isolated" eq. points
- 2: Finite escape time
- 3: Limit cycles
- 4: Chaos

Ex1



- If $\exists$ a neighborhood around $\bar{x}$ which does not contain any other eq.
 $\Rightarrow \bar{x}$ is an isolated eq.
- All neighborhoods around $\bar{x}$ contain at least one other eq.
 $\Rightarrow \bar{x}$ is not an isolated eq.